

STACK

Q6) Stock Span Problem

DS	Session	Stack	Question Information	Level 1	Challenge 46
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Question description

The **stock span problem** is a financial problem where we have a series of n daily price quotes for a stock and we need to calculate span of stock's price for all n days.

The span S_i of the stock's price on a given day i is defined as the maximum number of consecutive days just before the given day, for which the price of the stock on the current day is less than or equal to its price on the given day.

For example, if an array of 7 days prices is given as {100, 80, 60, 70, 60, 75, 85},

then the span values for corresponding 7 days are {1, 1, 1, 2, 1, 4, 6}

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CODE:

```

C main.c > main
1  #include <stdio.h>
2
3  void calculateSpan(int price[], int n, int S[]) {
4      int st[n], top = -1;
5      st[++top] = 0;
6      S[0] = 1;
7
8      for (int i = 1; i < n; i++) {
9          while (top >= 0 && price[st[top]] <= price[i]) {
10             top--;
11         }
12         S[i] = (top == -1) ? (i + 1) : (i - st[top]);
13         st[++top] = i;
14     }
15 }
16
17 void printArray(int arr[], int n) {
18     for (int i = 0; i < n; i++) {
19         printf(format: "%d ", arr[i]);
20     }
21     printf(format: "\n");
22 }
23
24 int main(void) {
25     int n;
26     if (scanf(format: "%d", &n) != 1) return 0;
27
28     int price[n], S[n];
29     for (int i = 0; i < n; i++) {
30         scanf(format: "%d", &price[i]);
31     }
32
33     calculateSpan(price, n, S);
34     printArray(arr: S, n);
35
36     return 0;
37 }

```

OUTPUT:

```
[suraj@suraj-83gs test]$ ./main
6
10 41 15 90 10 70
1 2 1 4 1 2
[suraj@suraj-83gs test]$
```